HERA HACKXX'26

24HRS WOMEN'S NATIONAL LEVEL HACKATHON

HERA-SIST-022 | SASTRA SLAYERS

Problem statement: Deepfake technology is increasingly being used to exploit women's identities, with limited detection and reporting infrastructure

Domain: Women's Safety & Security

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DEEPSAFE HER

Proposed Solution

- Web-based deepfake risk detection system
- Upload suspicious image/video
- Identify visual manipulation indicators
- Generate risk level (Low / Medium / High)
- Create structured case ID

How It Addresses the Problem

- Provides early-stage deepfake risk screening
- Builds detection + reporting infrastructure
- Preserves tamper-proof digital evidence
- Encourages safer digital interaction

Innovation

- Risk-based visual indicator model
- Simple, accessible, prevention-focused system
- Designed specifically for women's digital safety

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TECHNICAL APPROACH

TECHNOLOGIES USED:

- HTML, CSS, JavaScript
- Risk Scoring Algorithm
- Firebase Firestore (Evidence Vault)
- Timestamp-based Documentation

SYSTEM FLOW:

User Upload
→ Select Risk Indicators
→ Risk Score Calculation
→ Case ID Generation
→ Store in Firebase
→ Display Case History

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FEASIBILITY AND VIABILITY

ANALYSIS:
Modular architecture allows rapid model updates as deepfake generation evolves.

CHALLENGES:
- High computational cost of video processing
- Rapid advancement of anti-detection AI

STRATEGIES:
- Model quantization for optimized inference
- Logic-based filtering to reduce false positives
- Lightweight frame sampling instead of full video processing

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IMPACT AND BENEFITS

Potential Impact:
Empowers women with a specialized “first-response” digital verification tool.

Benefits:

- **Social:** Restores trust in digital identity and consent
- **Legal:** Structured metadata + case IDs streamline reporting
- **Technological:** AI-driven early detection + secure evidence preservation

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RESEARCH AND REFERENCES

AI Research:
"An Image is Worth 16x16 Words" – Vision Transformer (ViT), 2020
[2010.11929] An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

Datasets:
DeepFake Detection Challenge (DFDC)
Deepfake Detection Challenge | Kaggle
FaceForensics++ Benchmark

Statistics:
UN Women 2025/26 reports on synthetic media & gender-based online exploitation
Publications | UN Women – Headquarters

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